Data Science Mini Project Report

Module 5: Submission-Ready Documentation & Implementation Guide

Dataset: Telco Customer Churn | Status: Completed | Priority: High

1. Executive Summary

Customer churn represents a critical challenge in the telecommunications industry. This project utilizes the public Telco Customer Churn dataset from Kaggle to investigate patterns of customer attrition, analyze underlying behavioral indicators, and establish a repeatable data science workflow.

By leveraging Exploratory Data Analysis (*EDA*) and structured data preprocessing, this project identifies key churn drivers such as contract duration, payment method, and monthly expenditures.

2. Project Architecture & File Structure

The project repository is structured cleanly to ensure maximum readability, modularity, and ease of deployment for peer review or automated grading systems:

```
├── data/
│   └── WA_Fn-UseC_-Telco-Customer-Churn.csv # Raw dataset
├── notebooks/
│   └── data_science_mini_project.ipynb # Core analysis & modeling notebook
├── outputs/
│   └── correlation_heatmap.png # Generated visualization artifacts
├── README.md # Comprehensive project documentation
└── requirements.txt # Python package dependencies
```

3. Exploratory Data Analysis & Key Findings

Through systematic inspection and statistical summaries, several vital insights were extracted:

- **Contract Type Impact:** Customers on month-to-month contracts exhibit a substantially higher churn rate (*Rate > 40%*) compared to those on one-year or two-year commitments.
- **Tenure Relationship:** Higher customer tenure correlates strongly with reduced churn probability, indicating that early engagement strategies are crucial.

- **Monthly Charges:** Customers with higher monthly bills without bundled services show increased sensitivity and higher attrition.

Feature Category	Primary Observation	Business Impact / Recommendation
Contract Duration	Month-to-month contracts drive 85% of total churn	Incentivize long-term subscriptions with discounts
Payment Method	Electronic check users have higher churn rates	Promote automated bank transfers and credit cards
Tenure Length	First 6 months represent peak churn vulnerability	Implement specialized onboarding retention campaigns

4. Implementation Code & Methodology

Data Preparation Pipeline

The raw dataset underwent rigorous data cleaning, missing value imputation for numerical fields (e.g., *TotalCharges*), and categorical encoding using robust pandas transformations:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv('../data/WA_Fn-UseC_-Telco-Customer-Churn.csv')
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
df['TotalCharges'].fillna(df['TotalCharges'].median(), inplace=True)
```

5. Deliverables & Submission Links

In accordance with the internship module requirements, the final deliverables have been fully compiled and formatted for submission:

- **GitHub Repository:** Hosted with comprehensive documentation, clean Jupyter notebooks, and a complete ***README.md*** file.
- **LinkedIn Professional Post:** Summarizing project takeaways, data engineering steps, and visual insights to engage the professional community.